

SQL Constraint Scenario Questions with Answers

Question: 1. Employee Registration System

Use PRIMARY KEY with AUTO_INCREMENT (IDENTITY/SERIAL depending on DB). PRIMARY KEY ensures each EmployeeID is unique and not NULL. AUTO_INCREMENT automatically generates the next ID.

Question: 2. Online Banking Application

Use UNIQUE on Email. It prevents duplicate email addresses while allowing each customer to have a distinct login email.

Question: 3. College Management System

Use FOREIGN KEY on Student.DepartmentID referencing Department.DepartmentID. This enforces referential integrity and prevents assigning students to non-existent departments.

Question: 4. E-Commerce Order Management

Use CHECK (Price > 0). This rejects zero or negative prices and enforces the business rule.

Question: 5. Hospital Patient Registration

Use NOT NULL for PatientName so it is mandatory, and DEFAULT 'Unknown' for BloodGroup so that value is stored automatically when omitted.